**Numerical analysis of superconducting phases in the extended Hubbard model with non-local pairing**

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**Abstract**

[To be continued...]

- Appendix on $s$-wave SC ghost simulations on HM
- Appendix on triplet pairing in MFT self-consistent scheme
- Set ComputerModern font in all CairoMakie figures

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**List of symbols and abbreviations**

| | |
|---|---|
| AF | Antiferromagnetic |
| BCS | Bardeen-Cooper-Schrieffer (theory) |
| BZ | Brillouin zone |
| $\beta$ | Inverse temperature |
| $\Delta^{(\gamma)}$ | $\gamma$-wave component of the gap function |
| DoF | Degree of freedom |
| EHM | Extended Hubbard model |
| EV(s) | Expectation value(s) |
| FS | Fermi surface |
| HF | Hartree-Fock |
| HFP(s) | Hartree-Fock parameter(s) |
| HM | Hubbard model |
| $L_\ell$ | Number of lattice sites in direction $\ell$ |
| iHF | Iterative Hartree-Fock |
| LRT | Linear Response Theory |
| MBZ | Magnetic Brillouin zone |
| MFT | Mean-Field Theory |
| NBZ | Nematic Brillouin zone |
| NN(s) | Nearest neighbor(s) |
| RWC(s) | Relevant Wick's contraction(s) |
| RMP(s) | Renormalized model parameter(s) |
| SC | Superconductor |
| SDW | Spin-density wave |
| SSB | Spontaneous symmetry breaking |
| $T_\mathrm{c}$ | Critical temperature |

# Introduction

This thesis project is about my favorite ice cream flavor. [To be continued...]

## Chapter 1

# Introduction to the Extended Hubbard Model

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